Operating System (ICE 227)

Md. Azizul Hakim
Dept. of ICE, DIU.

Outline

- Operating System Structure

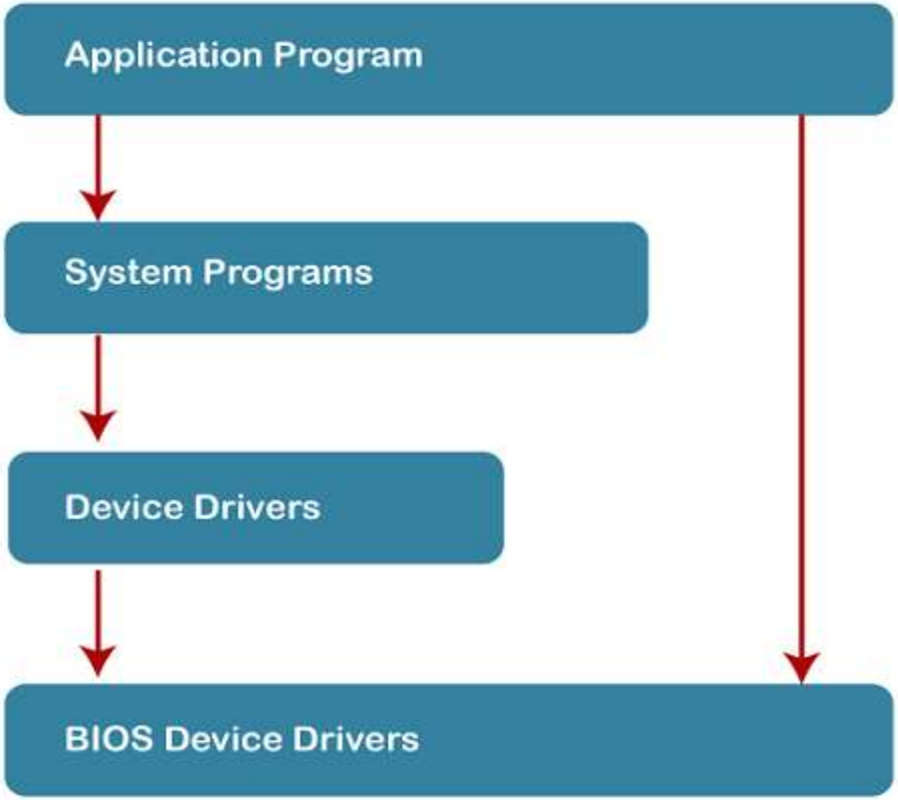
Operating System Structure

- Simple Structure
- Monolithic Structure
- Layered Approach Structure
- Micro-Kernel Structure
- Exo-Kernel Structure
- Virtual Machines

SIMPLE STRUCTURE

- It is the most straightforward operating system structure, but it lacks definition and is only appropriate for usage with tiny and restricted systems
- Four layers: ROM BIOS device drivers, MS-DOS device drivers, application programs, and system programs.

SIMPLE STRUCTURE



Simple Structure Diagram

SIMPLE STRUCTURE

Advantages:

- Because there are only a few interfaces and levels, it is simple to develop.
- Because there are fewer layers between the hardware and the applications, it offers superior performance.

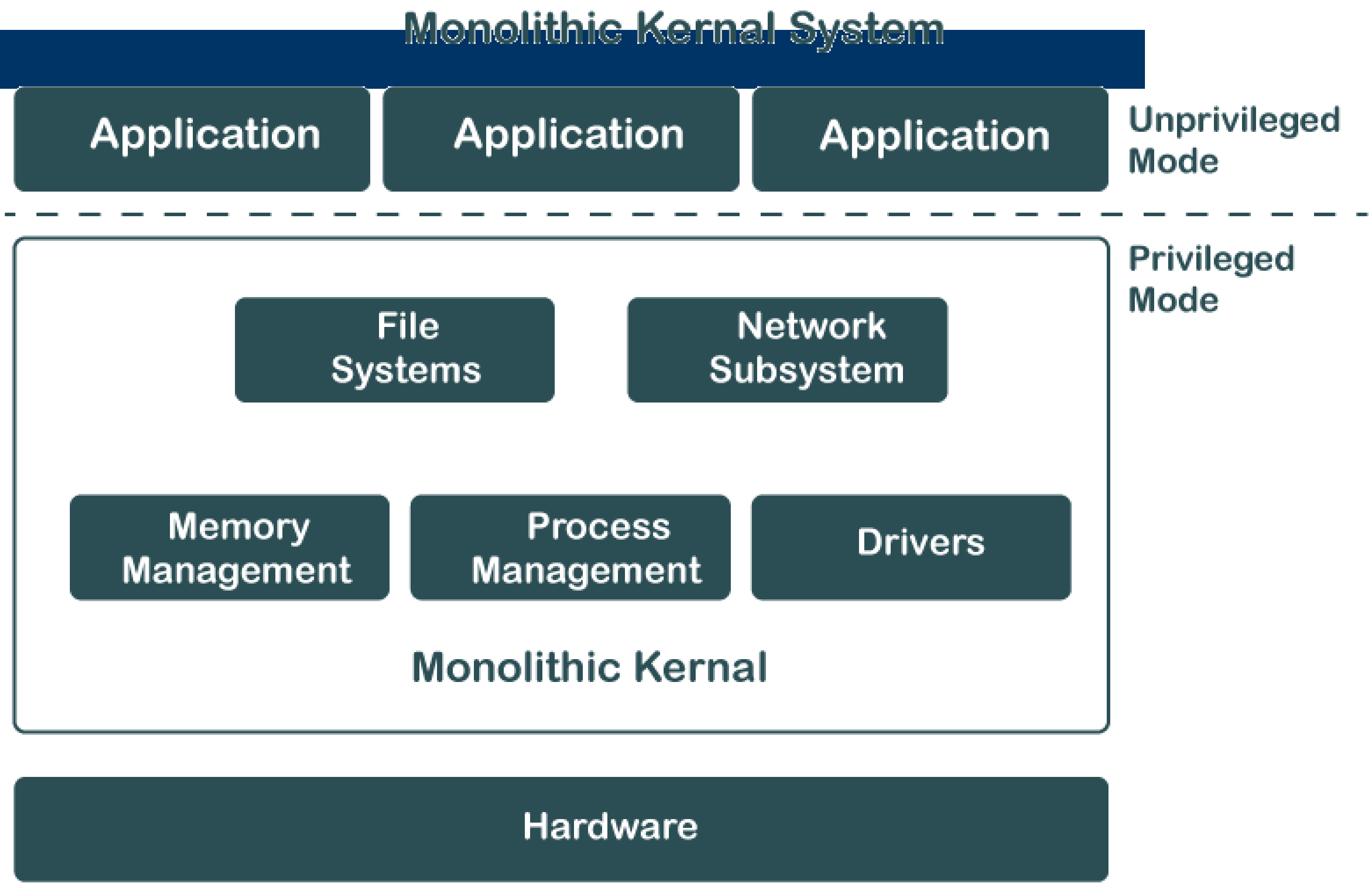
Disadvantages:

- The entire operating system breaks if just one user program malfunctions.
- Since the layers are interconnected, and in communication with one another, there is no abstraction or data hiding.
- The operating system's operations are accessible to layers, which can result in data tampering and system failure.

Monolithic STRUCTURE

- The monolithic operating system controls all aspects of the operating system's operation, including file management, memory management, device management, and operational operations.
- The monolithic operating system is often referred to as the monolithic kernel. Multiple programming techniques such as batch processing and time-sharing increase a processor's usability. Working on top of the operating system and under complete command of all hardware, the monolithic kernel performs the role of a virtual computer..

Monolithic STRUCTURE



Monolithic STRUCTURE

Advantages:

- Because layering is unnecessary and the kernel alone is responsible for managing all operations, it is easy to design and execute.
- Due to the fact that functions like memory management, file management, process scheduling, etc., are implemented in the same address area, the monolithic kernel runs rather quickly when compared to other systems. Utilizing the same address speeds up and reduces the time required for address allocation for new processes.

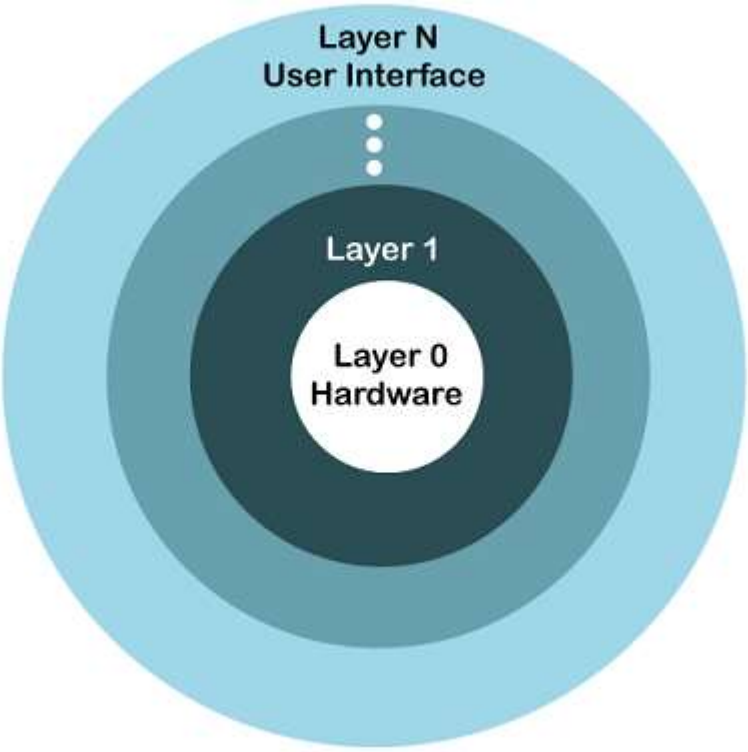
Disadvantages:

- The monolithic kernel's services are interconnected in address space and have an impact on one another, so if any of them malfunctions, the entire system does as well.

Layered STRUCTURE

- The OS is separated into layers or levels in this kind of arrangement. Layer 0 (the lowest layer) contains the hardware, and layer 1 (the highest layer) contains the user interface (layer N). These layers are organized hierarchically, with the top-level layers making use of the capabilities of the lower-level ones.
- The functionalities of each layer are separated in this method, and abstraction is also an option. Because layered structures are hierarchical, debugging is simpler, therefore all lower-level layers are debugged before the upper layer is examined.

Layered STRUCTURE



Layered Structure Diagram

Layered STRUCTURE

Advantages:

- Work duties are separated since each layer has its own functionality, and there is some amount of abstraction.
- Debugging is simpler because the lower layers are examined first, followed by the top layers.

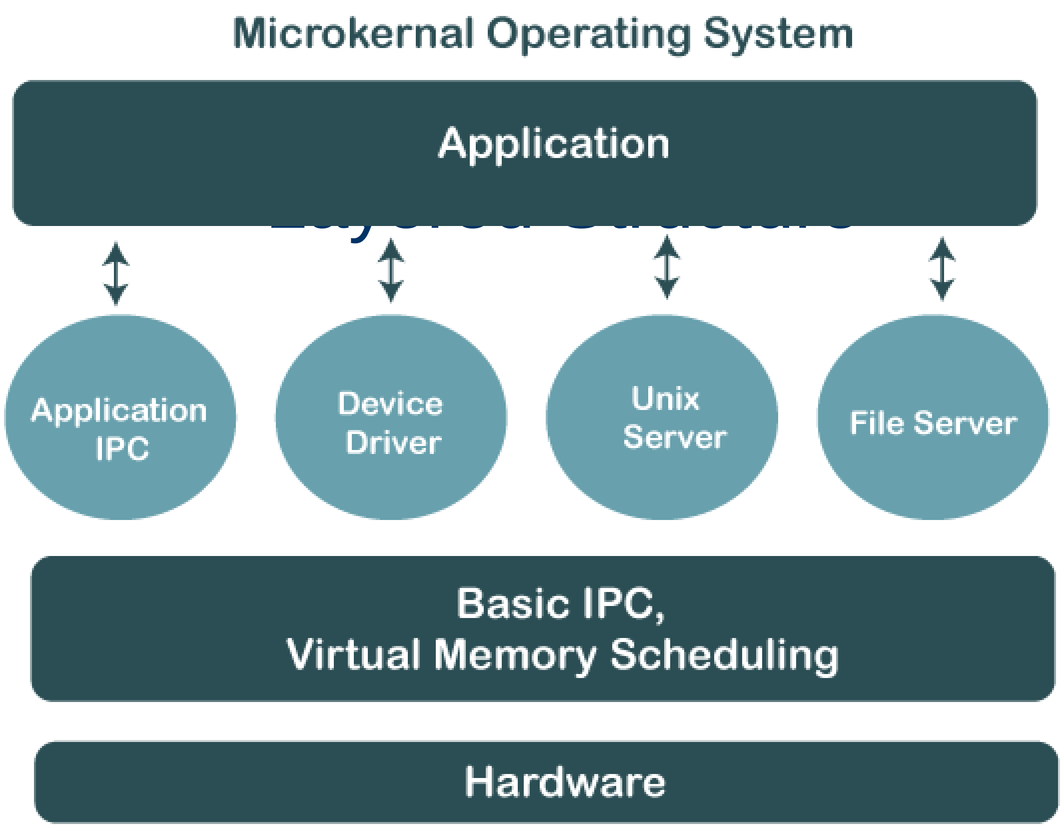
Disadvantages:

- Performance is compromised in layered structures due to layering.
- Construction of the layers requires careful design because upper layers only make use of lower layers' capabilities.

Micro-Kernel STRUCTURE

- The operating system is created using a micro-kernel framework that strips the kernel of any unnecessary parts. Systems and user applications are used to implement these optional kernel components. So, Micro-Kernels is the name given to these systems that have been developed.
- Each Micro-Kernel is created separately and is kept apart from the others. As a result, the system is now more trustworthy and secure. If one Micro-Kernel malfunctions, the remaining operating system is unaffected and continues to function normally.

Micro-Kernel STRUCTURE



Micro-Kernel STRUCTURE

Advantages:

- It enables portability of the operating system across platforms.
- Due to the isolation of each Micro-Kernel, it is reliable and secure.
- The reduced size of Micro-Kernels allows for successful testing.
- The remaining operating system remains unaffected and keeps running properly even if a component or Micro-Kernel fails.

Disadvantages:

- The performance of the system is decreased by increased inter-module communication.
- The construction of a system is complicated.

ExoKernel STRUCTURE

- An operating system called Exokernel was created at MIT with the goal of offering application-level management of hardware resources. The exokernel architecture's goal is to enable application-specific customization by separating resource management from protection. Exokernel size tends to be minimal due to its limited operability.
- The OS sits between the programs and the actual hardware, it will always have an effect on the functionality, performance, and breadth of the apps that are developed on it. By rejecting the idea that an operating system must offer abstractions upon which to base applications, the exokernel operating system makes an effort to solve this issue. The goal is to give developers as few restriction on the use of abstractions as possible while yet allowing them the freedom to do so when necessary

ExoKernel STRUCTURE

Advantages:

- Application performance is enhanced by it.
- Accurate resource allocation and revocation enable more effective utilisation of hardware resources.
- New operating systems can be tested and developed more easily.
- Every user-space program is permitted to utilise its own customised memory management.

Disadvantages:

- A decline in consistency
- Exokernel interfaces have a complex architecture.

Virtual Machine(VM)

- The hardware of our personal computer, including the CPU, disc drives, RAM, and NIC (Network Interface Card), is abstracted by a virtual machine into a variety of various execution contexts based on our needs, giving us the impression that each execution environment is a separate computer. A virtual box is an example of it.
- Using CPU scheduling and virtual memory techniques, an operating system allows us to execute multiple processes simultaneously while giving the impression that each one is using a separate processor and virtual memory. System calls and a file system are examples of extra functionalities that a process can have that the hardware is unable to give. Instead of offering these extra features, the virtual machine method just offers an interface that is similar to that of the most fundamental hardware. A virtual duplicate of the computer system underneath is made available

Virtual Machine(VM)

Advantages:

- Due to total isolation between each virtual machine and every other virtual machine, there are no issues with security.
- A virtual machine may offer an architecture for the instruction set that is different from that of actual computers.
- Simple availability, accessibility, and recovery convenience.

Disadvantages:

- Depending on the workload, operating numerous virtual machines simultaneously on a host computer may have an adverse effect on one of them.
- When it comes to hardware access, virtual computers are less effective than physical ones

Thank You